DESIGN REGISTRATION

PRELIMINARY INFORMATION FORM
(PLEASE TYPE OR PRINT LEGIBLY)

Full name and address of the applicant/s	1. Nivrutti Vishram Patil 2. Snehal Arun Sagare 3. Shubham Satish Modi 4. Aniket Raju Chhatre 5. Omkar Pradip Ozarkar 6. Prajakta Pralhad Patil 7. Shadab Ismail Pinjari 8. Akanksha Raosaheb Gite 9. Jay Sanjeev Jagtap 10. Sarthak Shashikant Tisgaonkar
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Category of applicant	Natural Person
GST No.	
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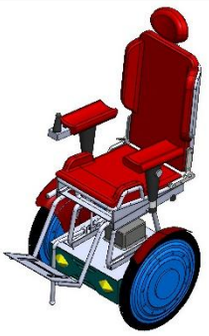
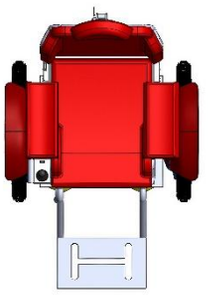
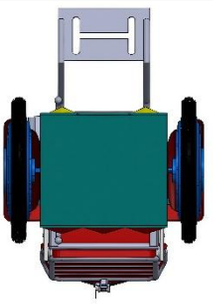
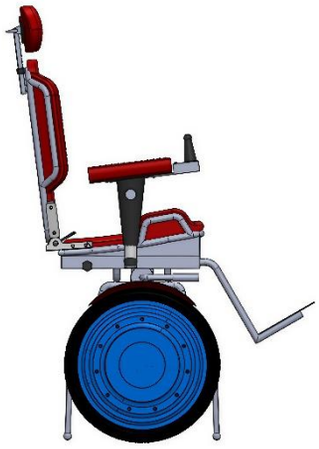
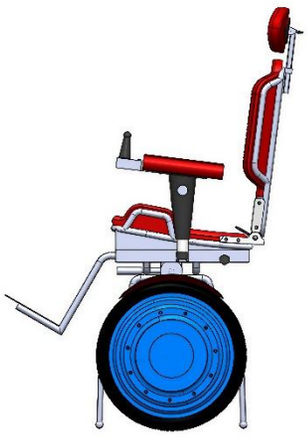
Name / Title or Description of the article to be protected in less than 15 words	Self-Balancing Multifunction Wheelchair
Description of article **(Max 200 words)**	The Self-Balancing Multifunction Wheelchair is an innovative personal mobility solution designed to enhance independence, safety, and comfort for individuals with physical disabilities. It addresses critical user-centric design challenges such as posture alignment and the reduction of musculoskeletal disorders (MSDs) caused by poor seating posture and prolonged strain. Based on the inverse pendulum principle, the wheelchair utilizes gyroscopic sensors, accelerometers, and a real-time feedback control system to maintain dynamic balance on two wheels. An intuitive joystick allows the user to move forward, backward, and rotate with minimal effort. The system features a reclinable anti-submarine seat, height-adjustable armrests, and an adjustable head restraint to provide ergonomic support and reduce fatigue. Powered by a lithium-ion battery, the system ensures reliable performance with efficient energy use. Its compact and modular design ensures smooth maneuverability in both indoor and outdoor environments, especially in tight urban spaces. The wheelchair includes a motorized drive system and onboard charging capability for user convenience. Designed in alignment with Indian anthropometric data, the product ensures better fit and usability for local users. This solution stands out as an accessible, intelligent, and user-friendly mobility aid that promotes dignity and autonomy. It represents a significant step toward inclusive assistive technology, offering a sustainable alternative to conventional and imported wheelchairs.
Please provide photographs or drawings of the article to be protected as a novel design.	Paste the appropriate views in the below mentioned fields

1. Perspective view	
2. Front view	
3. Rear View	
4. Top View	
5. Bottom View	

6. Left Hand Side View	
7. Right Hand Side View	
NOTE: 4 sets of photographs of each of the views (1-7) of the article should be provided for design registration application.	
Duly executed Power of Authority by the applicant or the authorized signatory. Please note that the Power of Authority is not required to be legalized or notarized.	

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